Firas Mostafa

Full-Stack & AI Engineer | Flutter · Django · Machine Learning

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SUMMARY

Software Engineer (2026 graduate, AI specialization) who builds complete applications end to end: cross-platform Flutter apps, Django REST and real-time ASGI backends, and the machine-learning and LLM systems behind them. Freelance experience extending a large third-party Flutter codebase for a client. Designed and built a multi-container AI health platform solo as a graduation thesis, from model training to a load-tested API and a Flutter client. Available for remote and on-site positions.

TECHNICAL SKILLS

- **Mobile:** Flutter, Dart, Flutter Web, BLoC / Cubit, GetX, Provider, Clean Architecture, Firebase (Auth, FCM, Firestore), Google Maps & Google Fit SDKs, responsive UI
- **Backend:** Python, Django REST Framework, Daphne (ASGI), FastAPI, Celery, Redis, WebSockets, JWT
- **AI & Machine Learning:** PyTorch, XGBoost, LightGBM, CatBoost, LangChain, Ollama, RAG, BGE-M3, pgvector, Faster-Whisper
- **Data, Infrastructure & Workflow:** PostgreSQL, Docker, Docker Compose, Git, GitHub Projects (Agile/Kanban), Postman, TDD

EXPERIENCE

Freelance Flutter Developer — Food Delivery Platform (Client Project) | *Flutter, GetX, Google Maps, Flutter Web*

Sep 2024 – Jan 2025 • Sole developer

- Customized a white-label multi-vendor food delivery platform (StackFood: customer, restaurant and delivery apps, built with Flutter and GetX) for a client, working within a large, undocumented third-party codebase.
- Restored the broken Flutter web app and got the other apps running on the web, reworking their layouts to be fully responsive across mobile, tablet and desktop.
- Upgraded the Flutter SDK and dependencies, fixed build and runtime issues, and deployed the web apps to Hostinger.

PROJECTS

Ursa — AI Health Assistant & Triage Platform (Graduation Thesis) | *Flutter, Django ASGI, PyTorch, LangChain, Docker*

Jan 2026 – Sep 2026 • Solo project • Demo video: youtu.be/WBz_ei6zKjU

- Architected a multi-container backend (Docker Compose): Django + Daphne ASGI gateway, Celery workers with Redis, PostgreSQL + pgvector, an Ollama LLM container, and Faster-Whisper ASR.
- Trained a 5-seed ResNet MLP ensemble on 246K Kaggle records across 721 disease classes (Top-1: 85.97%, Top-3: 95.88%) and a calibrated XGBoost + LightGBM + CatBoost severity classifier on 558K emergency triage records (ROC-AUC: 0.9295, Critical Recall: 94.00%).
- Built a medical RAG pipeline (BGE-M3 + pgvector over physician-verified Altibbi & MedQuAD corpora, Gemma 3-4B streamed over WebSockets) with a dual ML + LLM safety check that escalates high-risk cases.
- Built the Flutter app with Clean Architecture and BLoC: real-time AI chat, smartwatch vitals via the Google Fit SDK, habit analytics dashboards, FCM push notifications, biometric login, and PDF medical reports.
- Wrote 80 unit and integration tests (100% pass rate); load-tested at 87.72 req/s with 0% errors.

TasteCraft AI — AI Cooking & Nutrition Platform | *Flutter, Django REST, JWT, LangChain, Docker*

May 2025 – Sep 2025 • github.com/firas-mostafa/tastecraft-ai

- Built a full-stack platform for AI-driven recipe generation, calorie tracking, meal planning, and a voice-enabled chatbot with Arabic/English (RTL/LTR) support, on a Django REST + JWT backend built with TDD and Docker.

AgriSphere — Smart Agriculture & Crop Health App | *Flutter, REST APIs, Machine Learning*

Sep 2025 – Dec 2025 • Team project • github.com/firas-mostafa/agri_sphere

- Built the Flutter app for a team platform that helps farmers choose planting times and diagnose plant diseases with ML models, integrating REST APIs and dashboards in an Agile workflow.

EDUCATION

Al-Sham Private University — Damascus, Syria

B.Eng. in Informatics Engineering (AI Specialization) | 2021 – 2026

LANGUAGES

Arabic: Native • **English:** Professional working proficiency (strong written)